

(e) Chemical formulae, equations and calculations	
Students should:	
1.25	write word equations and balanced chemical equations (including state symbols): • for reactions studied in this specification • for unfamiliar reactions where suitable information is provided.
1.26	calculate relative formula masses (including relative molecular masses) (M_r) from relative atomic masses (A_r)
1.27	know that the mole (mol) is the unit for the amount of a substance
1.28	understand how to carry out calculations involving amount of substance, relative atomic mass (A_r) and relative formula mass (M_r)
1.29	calculate reacting masses using experimental data and chemical equations
1.30	calculate percentage yield
1.31	understand how the formulae of simple compounds can be obtained experimentally, including metal oxides, water and salts containing water of crystallisation
1.32	know what is meant by the terms empirical formula and molecular formula
1.33	calculate empirical and molecular formulae from experimental data
1.34C understand how to carry out calculations involving amount of substance, volume and concentration (in mol/dm³) of solution	
1.35C understand how to carry out calculations involving gas volumes and the molar volume of a gas (24 dm³ and 24 000 cm³ at room temperature and pressure (rtp))	
1.36	<i>practical: know how to determine the formula of a metal oxide by combustion (e.g. magnesium oxide) or by reduction (e.g. copper(II) oxide)</i>